Morales, Samantha
CSCI 3203: Tashfeen's Machine Learning I
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Homework 7

Question 1. $(b^{(\ell)}, \partial b^{(\ell)}/\partial \mathcal{E})$ must have the same dimensions. Show the following pairs have the same dimensions,

$$\left(b^{(2)}, \frac{\partial \mathcal{E}}{\partial b^{(2)}}\right), \quad \left(W^{(2)}, \frac{\partial \mathcal{E}}{\partial W^{(2)}}\right), \quad \left(b^{(1)}, \frac{\partial \mathcal{E}}{\partial b^{(1)}}\right), \quad \left(W^{(1)}, \frac{\partial \mathcal{E}}{\partial W^{(1)}}\right)$$

Because of the "rules for a three layered network" mentioned earlier, we said,

$$b^{(\ell)} = b^{(\ell)} - \eta \left(\frac{\partial \mathcal{E}}{\partial b^{(\ell)}}\right)$$

In order to do a matrix subtraction, both matrices must have the same dimensions. since η is just a scalar value, we know that $b^{(\ell)}$ and $\left(\frac{\partial \mathcal{E}}{\partial b^{(\ell)}}\right)$ must have the same dimensions. The same applies for $W^{(\ell)}$ and $\left(\frac{\partial \mathcal{E}}{\partial W^{(\ell)}}\right)$

Question 2. Use the chain rule to expand the following derivative,

$$\frac{\mathrm{d} f_n(f_{(n-1)}(f_{(n-1)}(\cdots f(x))))}{dx}$$

$$= \frac{\mathrm{d} f_n}{\mathrm{d} f_{n-1}} \times \frac{\mathrm{d} f_{n-1}}{\mathrm{d} f_{n-2}} \times \cdots \times \frac{\mathrm{d} f_1}{dx}$$

Question 3. Use the following recursive definition,

$$\frac{\partial \mathcal{E}}{\partial b^{(\ell)}} = W^{(\ell+1)^\dagger} \left(\frac{\partial \mathcal{E}}{\partial b^{(\ell+1)}}\right) \circ \sigma'(z^{(\ell)})$$

Okie, I did. ^w^

Question 4. Look up "Universal approximation theorem (UAT)" on the internet. Wikipedia is a good starting point. According to the UAT, multilayer feed-forward networks with as few as n hidden later(s) are universal approximators. What is the value of n?

$n = 1$

Question 5. Explore the Anderson's Iris data set. Take 15 setosas, 15 versicolors and 15 virginicas for your testing set. One-hot encode the training labels and you might want to use np.expand_dims(X, axis=-1) to turn the rows of a matrix X into column vectors. This is however dependant on your implementation.

(1) What is the baseline accuracy of the testing set?
$\frac{45}{150} = \frac{3}{10}$

(2) Draw a scatter plot of sepal width (cm) against the sepal length (cm). Colour each data point as setosa, versicolor, or virginica and label the legends appropriately.
Figure 1

(3) Write a neural network from scratch to classift the flowers. How did you pick the hyper parameters?
We picked them in class.

(4) Give the final values of the following,
   (a) Number of hidden layers and number of neural units in each hidden layer.
   1 Hidden Layer, 8 Neural Units.
   (b) The learning rate η.
   $\eta = 0.01$
   (c) The stopping criteria for training.
   The fixed number of iterations, 350 epochs.
   (d) The strategy for initialising the weights and biases
   Weights and Biases are initialized as zeros.

(e) The training error curve.
Figure 2

(f) The final testing accuracy.
88.89%

Submission Instructions

Submit a `data.py` with all your data processing and a `network.py` with all your neural network, training and testing code. Assure that Numpy and Matplotlib are the only external libraries in use. Finally submit a `<lastname>.pdf` file with all your answers and plots.

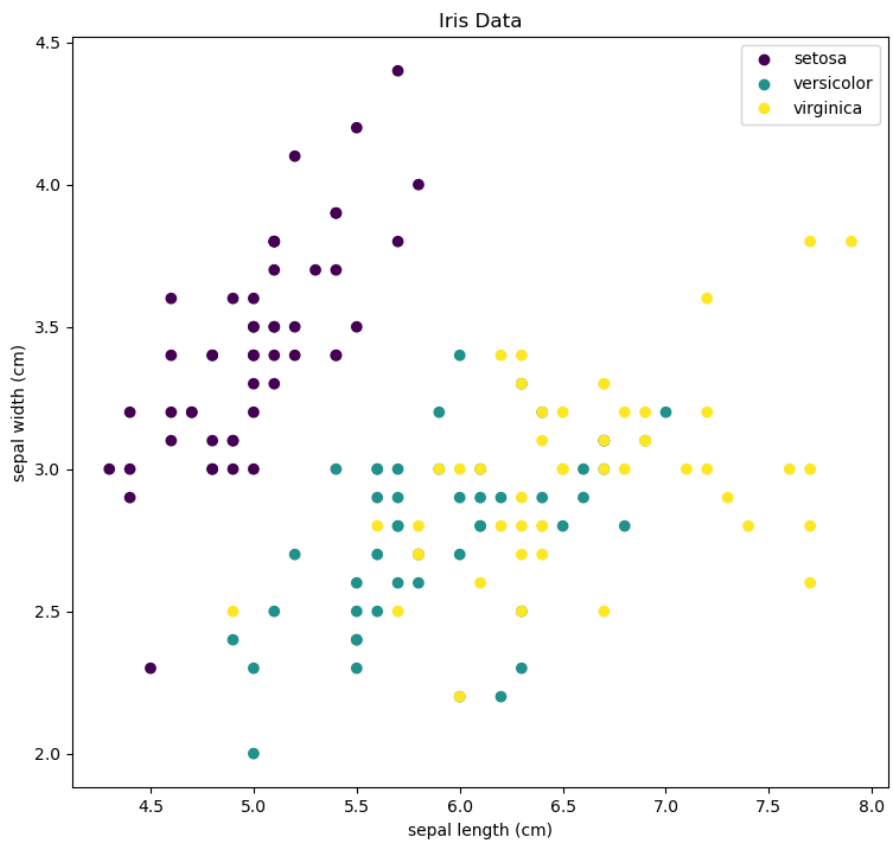

Figure 1. Scatter Plot

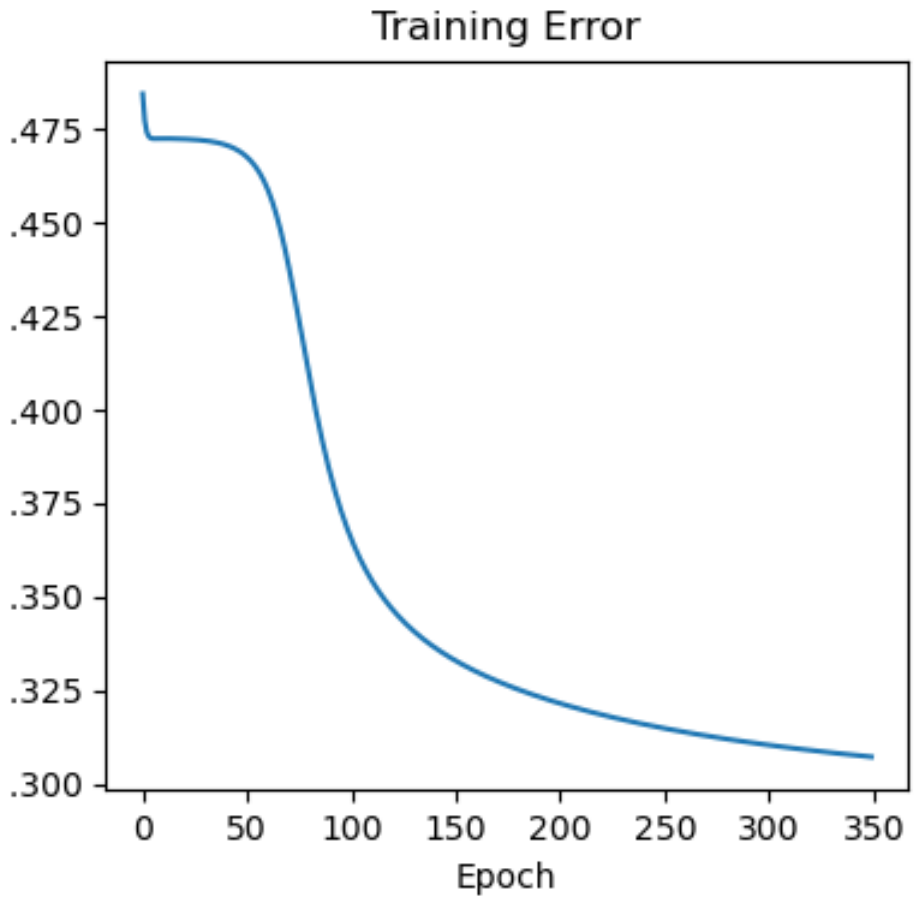

FIGURE 2. Training Error Curve